

Preparation, properties evaluation, and application in strawberry preservation of multi-functional arabica coffee husk pectin polysaccharide-cinnamaldehyde edible composite film

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Highlights

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Arabica coffee husk pectin polysaccharide-cinnamaldehyde films (CPF) were prepared.

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CPF possessed good mechanical properties, antioxidant and antibacterial activities.

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CPF films are environmentally sustainable and virtually non-toxic bioactivity films.

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CPF-1.5 film can extend the shelf life of strawberries by 4 days at 25 °C, 55–60 % RH.

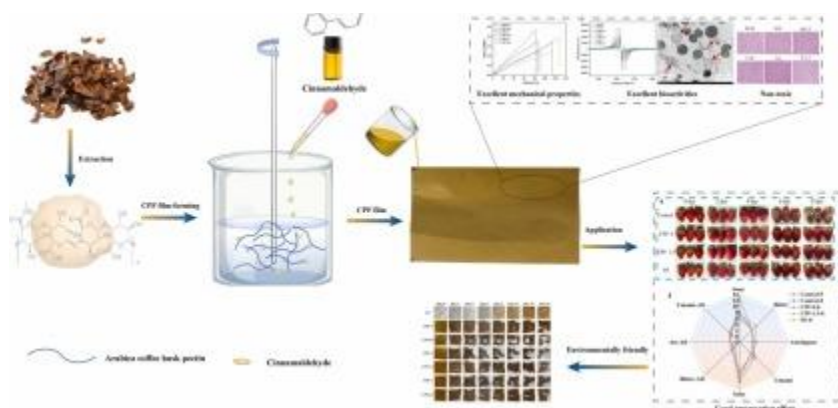
Abstract

With the growing global coffee consumption and production, the by-product coffee cherry husk has significantly increased, posing environmental concerns and challenges regarding its comprehensive utilization. Our previous study has shown that arabica coffee husks are rich in pectin polysaccharides (CP) and exhibits excellent film-forming properties.

Nevertheless, the CP application in encapsulating active compounds for preserving perishable fruits remains underexplored. This study focused on the development of films (CPF), which was loaded with different concentrations (0, 0.5, 1, 1.5, 2, and 2.5 %, v/v) of cinnamaldehyde (CIN) on the CP and their application in strawberry preservation. The results showed that all the CPF film-forming solutions possessed excellent film-forming

properties and antibacterial effects. CP and CIN jointly altered the microstructure of the CPF samples, enhancing their barrier properties, mechanical, antioxidant, and antibacterial properties. Among them, the CPF-1.5 group demonstrated the most outstanding comprehensive characteristics. The antibacterial mechanism experiments have confirmed that it could alter bacterial cell membrane permeability by binding to the amino acid residues of membrane receptors via CIN to form stable complexes. CPF films also exhibited exceptional biosafety and the ability to degrade in soil. In addition, it could extend the shelf life of strawberries by 4 days while preserving their color and firmness and enhancing their flavor profile (microbial load $< 1 \times 10^2$ CFU/g) at 25 °C and 55–60 % relative humidity. The findings demonstrated a dual advantage: it not only enhanced the comprehensive utilization of coffee by-products but also indicated that CP hold significant potential for preserving perishable fruits.

Graphical Abstract



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Introduction

Coffee, as one of the world's top three beverages, has experienced a continuous increase in both consumption and production due to its distinctive flavor and stimulating effects. Consequently, the rise in coffee by-products has become significant, presenting potential environmental challenges and necessitating comprehensive strategies for their utilization (Yu et al., 2025). Moreover, the temporary or permanent disposal of coffee husks in agricultural land may lead to significant environmental concerns, as it has the potential to inhibit plant root development and increase greenhouse gas emissions resulting from anaerobic decomposition (de Oliveira Fernandes et al., 2025). Coffee husks, as the primary by-product of coffee processing, constitute 40–50 % of fresh coffee cherries weight and represent the most abundant by-product generated during coffee processing (Fig. 1) (Li et

al., 2023). The coffee husks are rich in a variety of beneficial compounds, such as flavonoids, polyphenols, anthocyanins, and pectin polysaccharides. Notably, pectin polysaccharides constitute 20–30 % of the coffee husks; however, this valuable component has not been utilized effectively (Yu et al., 2025). This has resulted in a significant waste of resources, substantially decreasing the revenue of coffee enterprises and farmers, and hindering the high-value development of the coffee industry.

Pectin polysaccharides have demonstrated remarkable film-forming capabilities and have been utilized as carriers for a variety of active substances in fruit preservation. Zhang et al. (2025) proposed a strategy of interlocking symbiosis between the Pickering emulsion of nano-lignin and citrus essential oil and the pectin polysaccharide matrix interface, which can significantly extend the shelf life of kiwifruit and strawberries. Qian et al. (2025) developed a robust, antibacterial, and antioxidant chitosan-based composite film incorporating proanthocyanidins and carvacrol essential oil for strawberry preservation. The findings demonstrated that the film extended the shelf life of strawberries by 4 days and effectively mitigated weight loss (6.2 % of weight loss rates). Our previous study also found that the pectin polysaccharides extracted from Yunnan arabica coffee husks (CP) are a kind of polysaccharide composed of rhamnose, galacturonic acid, β -N-acetylglucosamine, glucose, galactose, xylose and arabinose linked by α -(1 $\rightarrow$ 6) glycosidic bonds. Moreover, its aqueous solution is a shear-thinning non-Newtonian fluid and exhibits excellent antioxidant properties, possessing the excellent performance required for film formation (Li et al., 2023). However, to our knowledge, there have been no reports so far on the production of CP as a fruit preservative film.

Therefore, this study focuses on the preparation of CP films (CPF) incorporated with varying concentrations of cinnamaldehyde (CIN). It systematically compares and analyzes their mechanical properties, physicochemical characteristics, antioxidant and antibacterial capabilities and underlying mechanisms, biosafety, and environmental friendliness. Finally, these films were used for the preservation of strawberries. This study aims to establish a foundation for the industrial utilization of coffee husks and their application in preserving perishable fruits.

Materials and chemicals

The mechanically dehulled sun-dried Arabica coffee husks were purchased from Baoshan Bitong Coffee Co., Ltd., (China). Extraction of pectin polysaccharides from coffee husks (CP) based on our previous study (Li et al., 2023). Sodium alginate was purchased from Qingdao Mingyue Seaweed Group Co., Ltd., (China). Cinnamaldehyde (CIN) (batch number: C805–630, purity $\geq$ 99 %) obtained by supercritical extraction of dried cinnamon bark, which was purchased from Guangzhou Jingjing Biotechnology Co.,

Characterization and antibacterial activity of CPF-forming solutions

A stable and uniformly dispersed forming solution is a prerequisite for the production of a functional composited film. And, the stability of the forming solution is determined by the droplet size, the distribution of electric potential, and the size of the polymer dispersed phase (Zhong et al., 2024). As shown in Fig. 2A, with the increase of CIN concentration, the CPF-forming solutions droplet size first decreased and then increased, with the smallest droplet size distribution occurring at

Conclusion

In this study, a preservative film for perishable fruits (strawberries) was successfully developed using CP and loaded with CIN. The prepared film-forming solutions (CPF-0, 0.5, 1.0, 1.5, 2.0, and 2.5) exhibited excellent dispersion stability, remarkable antibacterial properties, and non-Newtonian fluid behavior. After prepared into films, it was indicated that all of them had excellent properties of a preservative film (such as low CO₂ permeability and WVP, etc.) with a smoother surface, and

CRedit authorship contribution statement

Zelin Li: Writing – review & editing, Writing – original draft, Project administration, Funding acquisition. **Jiashun Gong:** Writing – review & editing, Supervision, Resources, Project administration. **Xiuwei Liu:** Validation, Investigation. **Lijuan Yu:** Project administration, Data curation. **Yifan Hu:** Investigation. **Hao Tian:** Writing – review & editing. **Di Wang:** Formal analysis, Data curation. **Jing Yan:** Methodology. **Chunyan Zhao:** Writing – original draft, Validation, Methodology, Data curation. **Ziqi**

Declaration of Competing Interest

The authors confirm that they have no conflicts of interest with respect to the work described in this manuscript.

Acknowledgements

This research were supported by Introduction and Development of Talents Program of Yunnan Academy of Agricultural Sciences (2024RCYP-14), Yunling Scholar Program (YNWR-YLXZ-2018–026), High-level Science and Technology Talent and Innovative Team Selection Program (202105AD160024), Yunnan Provincial Expert Li Puwang Workstation (202405AF140080), and Modern Agricultural Product Processing Technology Support Program.

Conflict of interest

The authors declare no conflict of interest.

